
DATA ACADEMY HACKATHON PROJECT

NOVA RETAIL

Technical Documentation & Gold ERD

End-to-end data analytics solution from source data preparation to an interactive Power BI reporting layer.

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The Information Lab | Hackathon Delivery

1. Project Overview and Architecture

Nova Retail is an end-to-end data analytics project developed for a global technology and electronics retailer. The objective is to transform heterogeneous source data into a reliable, tested and analytics-ready model that supports business intelligence reporting and sales performance analysis.

The solution addresses the company's "As-Is" situation, where data was fragmented across messy local files managed manually by different departments. By implementing a Modern Data Stack using a Medallion Architecture (Bronze, Silver, Gold), the pipeline transitions the company to a single, traceable source of truth.

Local Files → Dataiku → Snowflake → dbt → Power BI

Layer	Technology	Purpose
Source	Local files	Original business data
Preparation	Dataiku	Source preparation and ingestion
Bronze	Snowflake	Raw or minimally prepared source data
Silver	dbt	Cleaning and standardization
Gold	dbt	Dimensional modelling and business logic
BI	Power BI	Business analysis and visualization

Architecture principles

- **Separation of responsibilities:** Each technology in the stack has a strictly defined and scoped role to prevent overlapping logic.
- **Layered transformation:** Following the Medallion architecture, source data is progressively cleaned, standardized, and modeled for consumption.
- **Source preservation:** Data-quality issues are documented and handled gracefully rather than silently removed, ensuring auditability.
- **Traceability:** dbt source () and ref () dependencies provide explicit lineage from raw data to final BI models.
- **Transparent assumptions:** Known limitations in the source data (such as missing regions in sales) are documented explicitly.

2. Data Sources and Preparation

The pipeline integrates four source domains with different structures and business purposes. A crucial part of the process involves handling the severe data quality challenges present in the raw files.

Source	Format	Purpose	Main preparation
raw_sales_transactions.csv	CSV	Order-level sales	Dates standardization, customer info parsing, price cleaning, quantity handling
web_reviews.json	JSON	Customer reviews	Nested object flattening, UNIX timestamps conversion, ratings validation
finance_targets_2023.xlsx	XLSX	Monthly targets	Wide-to-long (Unpivot) transformation
ECB historical reference rates	CSV	FX conversion	2023 filter, USD/GBP to EUR preparation

Dataiku Preparation (Ingestion to Bronze)

Dataiku acts as the controlled entry point between the original local files and the Snowflake Bronze layer. The preparation layer performs source inspection, initial formatting, and ingestion preparation while intentionally keeping the detailed analytical transformations in dbt.

- **Sales Transactions:** The denormalized CSV is ingested preserving its original structure, allowing downstream dbt models to handle dirty formats.
- **Web Reviews:** The nested JSON is ingested. Dataiku ensures nested arrays/objects are properly structured for Snowflake variant columns or flattened prior to load.
- **Finance Targets:** The wide-format Excel/CSV file remains in its original structure (Jan-23 to Dec-23 columns) during ingestion, deferring the UNPIVOT operation to the SQL Silver layer.
- **ECB FX Rates:** The historical reference dataset is filtered to the 2023 reporting period and prepared for ingestion to support currency standardization.

3. Snowflake Data Warehouse

Snowflake provides the central cloud data warehouse for the project. It leverages the separation of storage and compute, organizing the data into a dedicated database and separate schemas for ingestion, transformation, and analytical consumption.

Object	Configuration
Database	NOVA_RETAIL
Schemas	BRONZE SILVER GOLD
Warehouse	NOVA_RETAIL_WH
Warehouse size	X-SMALL
Auto suspend	60 seconds
Auto resume	Enabled

Layer Definitions

Bronze Layer

Bronze contains the raw, prepared source datasets loaded directly from Dataiku. It acts as the immutable, stable input layer for all downstream dbt models.

```
NOVA_RETAIL.BRONZE.NOVARETAIL_RAW_SALES_TRANSACTIONS_COPY
NOVA_RETAIL.BRONZE.NOVARETAIL_WEB_REVIEWS_COPY
NOVA_RETAIL.BRONZE.NOVARETAIL_FINANCE_TARGETS_2023_COPY
NOVA_RETAIL.BRONZE.NOVARETAIL_EUROFXREF_HIST_PREPARED_COPY
```

Silver and Gold Layers

The **SILVER** schema contains cleaned and standardized source-level models materialized via dbt. The **GOLD** schema houses the final dimensional (Star Schema) and fact models specifically designed for BI consumption.

4. dbt Transformation Architecture

dbt (Data Build Tool) is responsible for the core transformation, modelling, testing, and documentation of the pipeline. The project strictly separates source-level transformations (Staging) from business-oriented analytical models (Core).

Layer	Models	Materialization	Role
Staging (Silver)	stg_sales_transactions stg_web_reviews stg_finance_targets stg_fx_ecb	View	Clean, cast data types, handle errors, and standardize raw source data
Core (Gold)	dim_date dim_customer dim_product fct_sales fct_targets	Table	Finalized, business-ready dimensional model

Staging Transformations (Bronze → Silver)

- **stg_sales_transactions:** Standardizes dates, splits the customer_info string by pipes into Name, Email, and Phone, identifies currencies, cleans price strings to numerics, normalizes discounts, and flags negative quantities as returns.
- **stg_web_reviews:** Trims text, converts mixed 10-digit and 13-digit UNIX timestamps into standard datetime formats, and validates ratings.
- **stg_finance_targets:** Transforms wide monthly columns (Jan-23 to Dec-23) via an UNPIVOT operation into a normalized structure: Region, Category, Month, TargetAmount.
- **stg_fx_ecb:** Reshapes ECB wide data into a Date + Currency + Rate structure for joining.

Core Transformations (Silver → Gold)

- Core models exclusively use `ref()` to reference staging models, preserving complete dbt lineage.
- Dimensions are built to provide descriptive filtering context, while fact tables contain measurable business events.
- Core models are materialized as tables to ensure optimal performance when queried by the Power BI Import engine.

5. Gold Data Model

The Gold layer follows a dimensional star-schema structure, with normalized dimensions and fact tables designed for BI consumption.

Model	Type	Grain	Purpose
DIM_DATE	Dimension	One row per calendar date	Calendar, month, week and Time Intelligence filtering
DIM_CUSTOMER	Dimension	One row per normalized customer email	Customer attributes and analytical key generation
DIM_PRODUCT	Dimension	One row per product ID/ category	Product and category drill-down analysis
FCT_SALES	Fact	One row per order	Revenue, profit, quantity and FX converted metrics
FCT_TARGETS	Fact	Region + Category + Month	Monthly financial targets for comparative analysis

Dimension Details

- **DIM_DATE:** Provides one row per calendar day for the 2023 reporting period. Key attributes include `date_day`, `year`, `month_number`, `month_name`, `quarter`, and various start-of-period markers required for DAX Time Intelligence.
- **DIM_CUSTOMER:** Customer keys are generated deterministically from normalized customer emails: whitespace is trimmed, the email is converted to lowercase, and an MD5 hash is used to create `customer_key`.
- **DIM_PRODUCT:** Contains `product_key`, `product_id`, and `product_category`. It directly supports the required Product Category → Product drill-down interactivity in Power BI.

Fact Details

- **FCT_SALES:** The sales fact has order-level grain. It contains transactional attributes alongside calculated revenue, profit, and standardized EUR conversion fields.
- **FCT_TARGETS:** The target fact has a coarser grain (Region + Category + Month) and contains the monthly `target_amount`. Target dates are aligned using the first day of each respective month.

6. Business Logic and FX Standardization

Revenue and profit metrics are calculated at the transaction level within the dbt Gold models before being aggregated by the BI layer.

```
Revenue = Price * Quantity * (1 - Discount)
Profit = Revenue * 0.30
```

Note: The uniform 30% margin on price is a mandated project business assumption utilized to calculate Total Profit.

Currency Handling Strategy

To provide accurate performance metrics to C-Level executives, all sales are standardized to a single currency (EUR).

Original Currency	Treatment in Model
EUR	EUR conversion rate = 1
USD	Converted using historical ECB reference rate
GBP	Converted using historical ECB reference rate
UNKNOWN	Retained in source currency, EUR fields remain NULL

For USD and GBP transactions, the dbt model joins the sales facts with the ECB FX reference model to select the latest available ECB reference rate **on or before** the transaction date. This prevents look-ahead bias (applying a future rate to a historical transaction).

If the transaction date is missing, or no applicable ECB rate exists in the loaded 2023 reference data, the revenue_eur and profit_eur fields remain NULL. Following strict data quality principles, no FX rate is artificially estimated or imputed.

7. Power BI Reporting Layer

Power BI serves as the Business Intelligence consumption layer, connecting strictly to the five Gold tables via Import mode.

Model Relationships

Dimension	Fact	Key	Cardinality & Direction
DIM_DATE	FCT_SALES	date_day	1 : Many (Single)
DIM_CUSTOMER	FCT_SALES	customer_key	1 : Many (Single)
DIM_PRODUCT	FCT_SALES	product_key	1 : Many (Single)
DIM_DATE	FCT_TARGETS	date_day	1 : Many (Single)

Module A: Executive Summary & KPIs

The semantic model handles the aggregation of row-level fact data into executive KPI metrics:

```
Total Revenue = SUM(FCT_SALES[REVENUE_EUR])
Total Profit = SUM(FCT_SALES[PROFIT_EUR])
Total Orders = DISTINCTCOUNT(FCT_SALES[ORDER_ID])
Average Order Value (AOV) = DIVIDE([Total Revenue], [Total Orders])
```

Time Intelligence

Users can select a reporting month via a global Date filter. The model uses DAX Time Intelligence functions to calculate Month-to-Date (MTD) vs Previous MTD, displaying percentage variances colored red/green.

Data Coverage Clarification: Because the provided raw data covers 2023 only, Previous YTD and YTD Variance % cannot be calculated from an actual prior-year dataset. The dashboard explicitly communicates that YTD prior-year comparisons are unavailable due to source data limitations.

Trend & Moving Average Analysis

The sales trend can be analyzed at Daily or Weekly granularity using a Power BI Field Parameter. A dynamic Moving Average Revenue measure responds dynamically to the selected number of days through a user-controlled parameter (configured from 3 to 30 days, default 7 days).

Module B: Target vs. Actuals

Using the Unpivoted Finance Targets, the dashboard compares Actual Revenue vs Target Revenue by Month. A visual indicator displays how far sales are from the monthly target. Interactivity is fully implemented, allowing drill-down from Category to Product.

Regional Analysis Clarification: Target Revenue is available by Region. However, Actual Revenue by Region cannot be calculated because `raw_sales_transactions.csv` lacks a region or country attribute. Therefore, Actual Revenue vs Target Revenue by Region is explicitly documented as unavailable at the sales level.

8. Data Quality and Validation

Data quality is rigorously addressed throughout the pipeline, from source preparation to transformation and model-validation stages via dbt tests.

Area	Identified Issue in Raw Data	Treatment Strategy
Sales Dates	Mixed formats, invalid/missing values, strings	Standardized to YYYY-MM-DD; unparseable values become NULL
Customer Info	Name, email and phone combined	Split into separate, dedicated fields in Staging
Price	Mixed currency symbols, spaces, separators	Currency symbol extracted; numeric price cleaned and cast
Quantity	Negative quantities present	Retained in facts but logically flagged as returns
Discount	Decimals, percentages, and 'N/A' strings	Standardized to decimal float; 'N/A' coerced to 0
Reviews	Mix of 10-digit and 13-digit UNIX timestamps	Length checked and converted to standard datetime format
Ratings	Contains 'null' strings, blanks, out-of-bounds	Values outside the 1–5 range are converted to NULL.
Targets	Wide horizontal monthly structure	UNPIVOT applied to achieve granular, queryable rows

dbt Testing Implementation: The pipeline utilizes generic dbt tests (`not_null`, `unique`, `accepted_values`, `relationships`) to ensure referential integrity between Dimensions and Facts. Singular tests are employed for complex business rules (e.g., ensuring `fct_targets` has a unique grain per Region/Category/Month).

8.1 Schema Drift Strategy

The pipeline separates source ingestion, cleaning, and analytical modelling specifically to reduce the impact of unexpected source changes (Schema Drift).

Source structures are inspected during the Dataiku ingestion/preparation stage before being loaded into Snowflake Bronze. dbt staging models then explicitly select and transform the fields required by the analytical layer, establishing a controlled boundary between raw source data and business models.

Unexpected changes, such as new columns being added by source systems, can therefore be detected at the ingestion level without automatically altering or breaking the Gold model.

Conversely, missing or renamed required columns would cause affected dbt models or tests to fail during the build phase, making the issue highly visible rather than silently propagating incorrect data to the BI layer.

9. Known Limitations and Assumptions

The following limitations are inherent to the supplied source data and are intentionally documented rather than hidden or artificially manipulated.

- **Regional actuals limitation:** The Module B requirement requests a comparison of Actual Revenue vs Target Revenue by Region. The Finance Targets dataset contains regional targets; however, the `raw_sales_transactions.csv` file does not contain a region or country attribute. Although customer country is available in the `web_reviews.json` source, using it to derive sales region would require an additional, undocumented business mapping (e.g., mapping country codes to EMEA, APAC, NA) and a risky assumption that review geography perfectly represents sales geography. Because this mapping was not provided by the business source, actual sales by region were not synthetically inferred to maintain data integrity.
- **Target Currency Assumption:** The currency of the finance targets is not explicitly specified by the source and is assumed to be EUR for comparison purposes. This is an analytical assumption and is not confirmed by the source dataset.
- **Missing Dates:** Seven sales records have missing or invalid transaction dates which cannot be resolved.
- **Unresolved Currencies:** Three sales records have UNKNOWN currency because the source text does not provide enough information to determine the symbol.
- **Missing FX Rates:** Eight sales records do not have EUR-converted revenue due to the absence of an applicable ECB rate within the loaded 2023 reference dataset.
- **Target Regions:** Four finance target records have Region assigned as NA (Null) based on the source structure; this is retained as-is.
- **Fact Grains:** FCT_TARGETS operates at a monthly aggregate grain, while FCT_SALES operates at a detailed order-level grain.
- **Direct Relationships:** FCT_SALES and FCT_TARGETS are not directly related. FCT_TARGETS is connected to DIM_DATE only. Product Category and Region are target-side attributes within FCT_TARGETS and do not create a relationship with DIM_PRODUCT.

Interpretation rule: Financial analysis should rely on EUR-converted measures where available. NULL EUR conversions are retained as traceable data-quality outcomes rather than being estimated via averages.

10. Gold ERD (Entity-Relationship Diagram)

The Gold layer follows a dimensional star-schema structure. Dimensions filter their corresponding fact tables using one-to-many relationships.

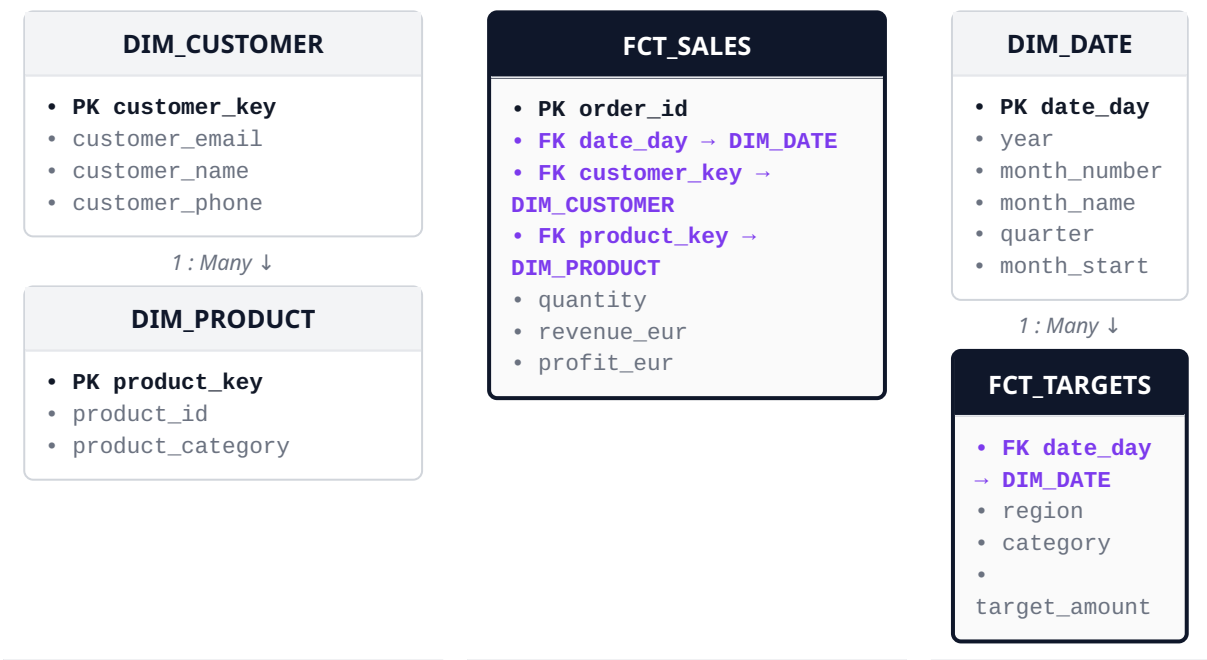


Diagram Key: PK = Primary Key | FK = Foreign Key | Cardinality = 1 : Many (Single Direction)
Note: FCT_SALES and FCT_TARGETS share DIM_DATE as their only common relationship path.

11. Deliverables & Final Architecture Summary

The project documentation and dbt analytical code are maintained and version-controlled.

Deliverable	Status / Content Provided
Technical Documentation (PDF)	End-to-end architecture, transformations, modelling logic, and assumptions.
Data Quality Log	Identified issues, cleaning rules, schema drift strategy, and limitation handling.
Gold ERD	Visual Entity-Relationship Diagram of the final dimensional layer.
Code Repository	dbt models, tests, macros, showcasing the pipeline from Bronze to Gold.
BI File (.pbix)	Interactive Power BI dashboard built strictly on top of the Gold dimensional model.

Final Outcome

Nova Retail now possesses a highly traceable, end-to-end analytics pipeline from fragmented, heterogeneous source data to a fully tested dimensional model and interactive business intelligence dashboard.

The Medallion architecture successfully separates ingestion, storage, transformation, modelling, and reporting responsibilities. By explicitly documenting source-data limitations (such as missing sales regions) and analytical assumptions (such as the 30% margin), the resulting End-to-End Modern Data Stack provides reliable, actionable metrics for C-Level executives without obscuring critical data quality realities.

Data Quality → Robust Modelling → Actionable BI

12. Orchestration — Exploratory Setup

An optional orchestration workflow was explored to evaluate how the Nova Retail pipeline could be triggered across Dataiku and dbt Cloud.

Dataiku & dbt Cloud

A Dataiku Scenario named Nova Retail – Daily Pipeline was configured with two sequential steps:

1. **Build Bronze Sources.** Builds the four Bronze datasets prepared through Dataiku: sales transactions, web reviews, finance targets and ECB FX reference data.
2. **Trigger dbt Build.** Calls the configured dbt Cloud API job to launch dbt build, with the intended flow continuing from Bronze into the dbt Silver and Gold layers.

Power BI Desktop & Service

The Gold-layer dashboard was completed successfully in Power BI Desktop and published to Power BI Service, where the semantic model was created. A further attempt was made to configure the Snowflake cloud connection and scheduled refresh. The Service connection could not be validated successfully, so automated refresh was not established.

Validation status

Exploratory only. The Dataiku/dbt workflow was configured but not executed end-to-end, and the Power BI Service refresh was not successfully validated. These experiments are therefore documented as extensions and are excluded from the validated production architecture.

Validated architecture

The validated project remains: Dataiku > Snowflake Bronze > dbt Silver > dbt Gold > Power BI. No validated data model, business logic, quality control or Gold schema was changed by these exploratory attempts.